Theory and Methods in Causal Inference

Lecture Notes

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STAT 5900B

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Preface

These lecture notes accompany the graduate course **STAT 5900B: Theory and Methods in Causal Inference** at the Department of Statistics, Iowa State University.

The course targets graduate students in statistics. It assumes familiarity with probability theory, mathematical statistics, and linear models at the level of a graduate sequence, but no prior exposure to causal inference.

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Organization

The notes are organized in four parts.

Part I — Foundations (Chapters 1–4) builds the graphical and conceptual foundation: the causal trinity (SEM, DAG, potential outcomes), DAGs and d-separation, the do-calculus and identification criteria, and the connection between potential outcomes and the do-calculus.

Part II — Identification (Chapters 5–8) bridges the graphical framework and observed data: randomization and back-door adjustment, propensity score methods, instrumental variables, and mediation and front-door identification.

Part III — Estimation (Chapters 9–12) covers semiparametric efficiency theory: estimating equations and influence functions, doubly robust estimation, flexible nuisance estimation with cross-fitting, and estimation under instrumental variables.

Part IV — Policy Learning and Sequential Decision Making (Chapters 13–15) covers policy evaluation, optimal policy learning, and dynamic treatment regimes.

Notation

Throughout these notes:

- $\mathrm{do}(T = t)$ denotes the do-operator (intervention).
- $Y(t)$ denotes the potential outcome under $T = t$.
- $\mathbb{E}[\,\cdot\,]$ denotes expectation; $\perp\!\!\!\perp$ denotes conditional independence.
- $\mathcal{G}$ denotes a DAG; $\mathcal{G}_T$ denotes the mutilated graph after intervening on T.

Chapter 1

Introduction

Learning Objectives

By the end of this chapter, students should be able to:

1. Articulate the distinction between an interventional distribution $f(y \mid \mathrm{do}(T{=}t))$ and an observational conditional distribution $f(y \mid T{=}t)$, and explain why they differ whenever T is endogenous.
2. Describe the same causal question in all three languages — SEM, DAG graph surgery, and potential outcomes — and begin to translate between them.
3. Derive the two densities explicitly in the Gaussian linear confounded model, and identify the endogeneity bias $\alpha\gamma\sigma_U^2/\sigma_T^2$ as the gap between them.
4. State the two-step paradigm (identification then estimation) and locate each step in the causal inference pipeline.
5. Distinguish the IV DAG from the mediation DAG and explain why conflating an instrument with a mediator leads to identification failure.

This chapter is an orientation, not a complete treatment. It introduces four ideas — the interventional/observational distinction, the causal trinity, the Gaussian confounded model as a running example, and the identification-versus-estimation paradigm — that will be developed rigorously in Chapters 2–4 and beyond. After one reading, students should have a map of the territory; the details will become clear as the course proceeds.

1.1 Motivation: Why Causal Inference Is Hard

Causal inference studies when and how the observational distribution can be used to recover the distribution that would arise under interventions.

1.1.1 The Central Question

Causal inference is concerned with a deceptively simple question: what would happen to Y if we were to *set* $T = t$? This is fundamentally different from the question that standard statistical procedures answer: what is the distribution of Y among units *observed* to have $T = t$?

Definition: Interventional vs. Observational Distribution

- The *interventional distribution* $f(y \mid \mathrm{do}(T{=}t))$ is the distribution of Y in a hypothetical world where T has been *externally set* to t, regardless of the usual data-generating mechanism for T.
- The *observational conditional distribution* $f(y \mid T{=}t)$ is the distribution of Y among the subpopulation of units for whom $T = t$ was *observed* to hold.

These two distributions coincide only when T is randomized (exogenous). In all other cases they differ, and the difference is *endogeneity bias*.

Definition: Exogeneity and Endogeneity

Suppose the outcome is generated by the structural equation

$$Y = f_Y(T, X, U),$$

where T is the treatment, X are observed covariates, and U represents unobserved common causes of T and Y (confounders).

- The treatment T is *exogenous given* X if $T \perp\!\!\!\perp U \mid X$.
- The treatment T is *endogenous given* X if $T \not\perp\!\!\!\perp U \mid X$, so that some unobserved variable U affects both T and Y.

Example: Drug Trial with Unmeasured Severity

Does a new drug ($T = 1$) reduce blood pressure (Y)? Suppose sicker patients are more likely to take the drug, so that unmeasured severity U is a common cause of T and Y. Then:

- *Observational study*: $\mathbb{E}[Y \mid T{=}1] > \mathbb{E}[Y \mid T{=}0]$. Drug users have higher blood pressure on average. A naïve analyst concludes the drug is harmful.
- *Randomized trial*: $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] < \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$. Forcing everyone to take the drug reduces blood pressure. The drug is beneficial.

Same data. Different questions. Opposite answers. The entire discrepancy is caused by the back-door path $T \leftarrow U \rightarrow Y$, which the observational quantity $\mathbb{E}[Y \mid T{=}1]$ cannot remove.

The Fundamental Problem of Causal Inference

We observe $f(y \mid T{=}t)$ directly from data. We want $f(y \mid \mathrm{do}(T{=}t))$. The goal of *identification theory* is to find conditions under which the former can be used to recover the latter.

1.2 The Causal Trinity: Three Languages for One Idea

The same causal question — what happens when we set $T = t$? — can be expressed in three distinct but equivalent languages. Collectively, we call these the **causal trinity**: the *structural equation model* (SEM), the *directed acyclic graph* (DAG) with graph surgery, and the *potential outcomes* framework. Each member of the trinity illuminates a different facet of the same underlying idea, and fluency in all three is essential for modern causal reasoning. Figure 1.1 depicts their relationships.

1.2.1 Language 1: Structural Equation Models

A structural equation model (SEM) represents the causal data-generating mechanism as a system of equations, one per variable, together with a joint distribution over exogenous errors.

Definition: Structural Equation Model (SEM)

A recursive SEM over variables (U, T, Y) is a system

$$U \sim p(u), \qquad T = f_T(U, Z, \delta), \qquad Y = f_Y(T, U, \varepsilon),$$

where δ and ε are mutually independent exogenous disturbances, Z is an observed instrument, and U is an unobserved confounder.

Remark: Error Independence in the NPSEM-IE

The assumption that the disturbances δ and ε are *mutually independent* is a substantive restriction, not merely a technical convenience. Because U is unobserved and enters both f_T and f_Y, error independence does not remove the dependence between T and $Y(t)$ induced by U; it restricts only the additional channel of dependence running through the equation-specific errors (δ, ε). Richardson

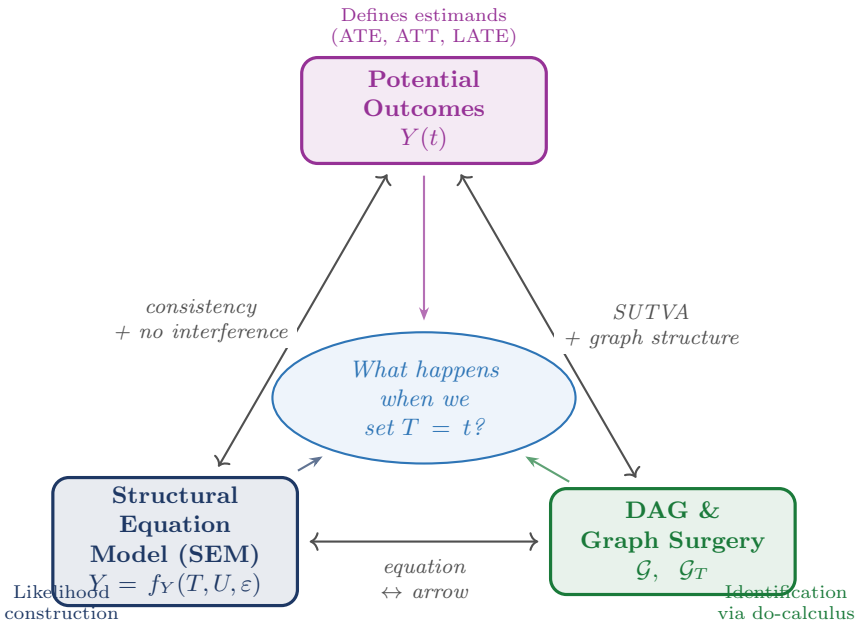

Figure 1.1: The **causal trinity**: three languages that express the same causal question from different vantage points. Potential outcomes define *what* we want to estimate; the DAG with graph surgery specifies *how* to identify it from observed data; the SEM provides the *generative mechanism* for likelihood construction. Double-headed arrows show the translation assumptions connecting adjacent languages. All three converge on the same central question (ellipse).

> and Robins (2014) call SEMs with this structure *Nonparametric Structural Equation Models with Independent Errors* (NPSEM-IEs).
> These notes adopt the NPSEM-IE framework throughout for two practical reasons: error independence ensures the DAG's Markov condition holds, making the Markov factorization of the joint density immediate; and it integrates cleanly with the graphical identification theory developed in later chapters.

Observational regime. T is determined by f_T. Because U appears in both f_T and f_Y, we have $\mathrm{Cov}(T, \varepsilon) \neq 0$ in general: this is endogeneity.

Interventional regime. The do-operator replaces the equation for T with a constant t, yielding the *mutilated system*:

$$U \sim p(u), \qquad T = t \text{ (fixed)}, \qquad Y = f_Y(t, U, \varepsilon).$$

Now T and U are independent by construction: the back-door path through U has been severed.

1.2.2 Language 2: Directed Acyclic Graphs

A DAG makes the same surgery graphically visible.

> Definition: Graph Surgery
>
> In the observational DAG, arrows into T encode the mechanisms that determine T (including back-door paths through unobserved variables). Under do(T=t), all arrows into T are *deleted*. The resulting mutilated graph $\mathcal{G}_T$ has no back-door paths from T to Y: the causal effect flows only along the direct path $T \to Y$.

The observational and mutilated DAGs for the Gaussian confounded model are shown side by side in Figure 1.3 in Section 1.4.

1.2.3 Language 3: Potential Outcomes

Definition: Potential Outcome [@rubin1974estimating]

$Y(t)$ is the outcome that *would have been observed* had T been set to t, possibly contrary to fact. **SUTVA** (stable unit treatment value assumption): $Y_i = Y_i(T_i)$, i.e., the observed outcome for unit i equals the potential outcome at the actual treatment received. This rules out interference between units and multiple versions of treatment.

Connection to the do-operator. Under standard structural assumptions (consistency + no interference), the potential outcome $Y(t)$ and the interventional distribution are linked by

$$Y(t) \;\sim\; f(y \mid \mathrm{do}(T{=}t)).$$

Remark

Potential outcomes define what we want to estimate (e.g., $\mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$), but they do not specify how to write an observed-data likelihood for estimation. The do-operator fills this gap by providing an explicit link between the interventional density and the observational distribution via identification rules (back-door, front-door, do-calculus).

1.2.4 Roadmap: How the Trinity Organizes These Notes

The three languages are not used equally throughout the course.

Part I — Foundations: the language of DAGs. The DAG and the do-operator are the primary language here because they make the interventional/observational distinction *syntactically explicit*: confounding is visible as an open back-door path, and identifying assumptions are visible as graph criteria.

Part II — Identification: bridging DAGs and observed data. Ignorability ($Y(t) \perp\!\!\!\perp T \mid X$) is the potential-outcomes counterpart of the back-door criterion; the propensity score $\pi(X) = P(T{=}1 \mid X)$ operationalizes it in observed data.

Part III — Estimation: semiparametric and machine-learning methods. The semiparametric efficiency framework — efficient influence functions, doubly robust estimators, double machine learning — is largely language-neutral: it works with whatever moment condition the identification step produces.

1.2.5 Historical Roots of the Causal Trinity

The three languages grew independently in different disciplines over more than a century, driven by different scientific questions and different communities. Understanding this history explains why the three schools sometimes use incompatible terminology for the same idea.

Structural Equation Models: genetics and econometrics. The oldest strand begins with the biologist Sewall Wright (1921), who introduced *path analysis* to decompose correlations among hereditary traits into direct and indirect causal contributions. The framework was transplanted into economics by Haavelmo (1943), who argued that economic relationships should be modeled as autonomous structural equations robust to policy interventions. The Cowles Commission formalized simultaneous equation systems throughout the 1940s–50s. Lucas's (1976) critique — that reduced-form equations break down when policy changes the data-generating process — is essentially a statement of the interventional/observational distinction, decades before that phrase existed.

Potential Outcomes: experimental statistics and epidemiology. Neyman (1923) introduced the notation $Y_i(t)$ in the context of randomized agricultural experiments. Fisher (1935) placed randomization at the foundation of causal inference. The crucial extension to *observational* studies was made by Rubin (1974), who formalized the assignment mechanism and stated SUTVA explicitly. Robins (1986) extended potential outcomes to longitudinal treatments and time-varying confounding via g-computation. Imbens and Angrist (1994) connected the framework to instrumental variables and defined the LATE.

DAGs and do-calculus: computer science and philosophy. Pearl developed Bayesian networks through the 1980s as a representation for probabilistic reasoning in AI. The pivotal step came in Pearl

(1995), where the *do-operator* was introduced as a formal syntax for distinguishing interventional from observational distributions within a graph-based algebra. The 2000 monograph *Causality* synthesised DAGs, the do-calculus, identification theory, and the connections to potential outcomes into a unified framework.

Convergence. By the 2000s it became clear that all three frameworks are, under standard assumptions, *logically equivalent*: any result expressible in one can be translated into the others. The causal trinity is therefore not three competing theories but three dialects of the same language, each optimized for a different task. Figure 1.2 summarizes the key milestones.

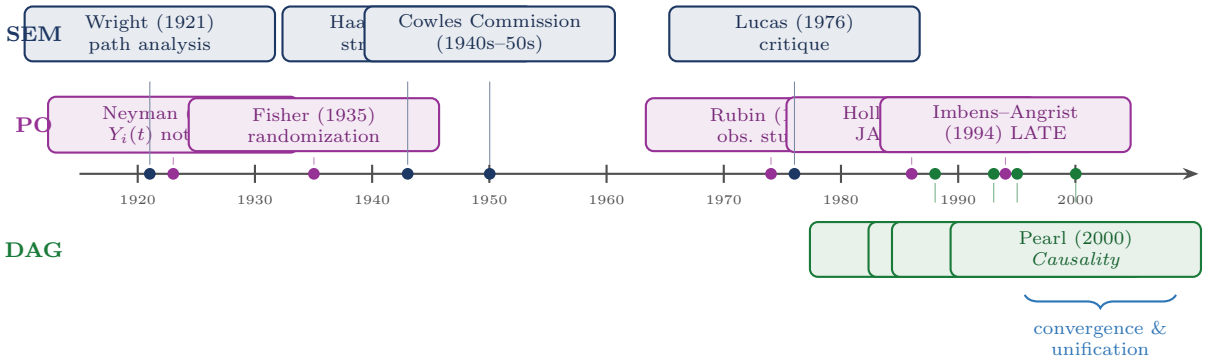

Figure 1.2: Key milestones in the three schools of causal inference. SEM (top, blue) originates in genetics and econometrics; PO (middle, purple) in experimental statistics; DAG (bottom, green) in computer science and philosophy. The three developed largely in isolation until converging in the 2000s.

1.3 The Core Distinction

The single most important inequality of this course is:

$$\underbrace{f(y \mid x, \mathrm{do}(T{=}t))}_{\text{interventional}} \neq \underbrace{f(y \mid x, T{=}t)}_{\text{observational}} \qquad \text{whenever } T \text{ is endogenous.} \tag{1.1}$$

The key assumption that connects the two densities is *conditional exchangeability*.

Definition: Conditional Exchangeability

Given observed covariates X and confounders U, the treatment assignment is *conditionally exchangeable with the intervention* if

$$Y(t) \perp\!\!\!\perp T \mid X, U,$$

or equivalently,

$$f(y \mid x, T{=}t, U{=}u) \;=\; f(y \mid x, \mathrm{do}(T{=}t), U{=}u).$$

This says that once we condition on all common causes (X, U) of T and Y, observing $T{=}t$ and intervening to set $T{=}t$ produce the same distribution of Y: there is no residual confounding given (X, U).

Under conditional exchangeability, the interventional density has a direct representation in terms of observable quantities.

Proposition: Back-Door Adjustment Formula

Under conditional exchangeability (Definition above),

$$f(y \mid x, \mathrm{do}(T{=}t)) \;=\; \int f(y \mid x, T{=}t, U{=}u)\, p(u \mid x)\, du. \tag{1.2}$$

Proof

Step (i): weight under the intervention. Under $\mathrm{do}(T{=}t)$, the arrow $U \to T$ is severed in the mutilated graph $\mathcal{G}_T$, so $U \perp\!\!\!\perp T \mid X$ under the intervention. The law of total probability therefore gives:

$$f(y \mid x,\, \mathrm{do}(T{=}t)) = \int f(y \mid x,\, \mathrm{do}(T{=}t),\, U{=}u)\; p(u \mid x)\; du.$$

Step (ii): replacing the interventional kernel. By conditional exchangeability, $f(y \mid x,\, \mathrm{do}(T{=}t),\, U{=}u) = f(y \mid x,\, T{=}t,\, U{=}u)$. Substituting completes the proof. □

Remark: Confounding Discrepancy

What OLS estimates is the observational density $f(y \mid x, T{=}t)$, not the interventional density. The gap

$$f(y \mid x,\, T{=}t) - f(y \mid x,\, \mathrm{do}(T{=}t)) = \int f(y \mid x,\, T{=}t,\, U{=}u)\big[\, p(u \mid x,\, T{=}t) - p(u \mid x)\,\big]\, du$$

is non-zero whenever U and T are dependent — the endogeneity condition.

1.3.1 Two Scenarios: Observed vs. Unobserved Confounder

Scenario 1: U is observed (no unmeasured confounding). When every component of U is recorded in the data, the back-door adjustment formula (Equation 1.2) becomes usable from data:

$$f(y \mid x,\, \mathrm{do}(T{=}t)) = \int f(y \mid x,\, T{=}t,\, U{=}u)\; p(u \mid x)\; du. \tag{1.3}$$

The confounder is averaged over its *population* distribution $p(u \mid x)$, not over the treatment-conditional distribution. A special case is *unconfoundedness*: if $U \perp\!\!\!\perp T \mid X$ already holds observationally, formula Equation 1.3 reduces to the familiar regression $\mathbb{E}[Y \mid X{=}x, T{=}t]$. This is the key assumption behind regression adjustment and propensity score methods studied in Chapter 6.

Scenario 2: U is unobserved (unmeasured confounding). When U is latent, formula Equation 1.3 is not usable. Additional structure beyond (Y, T, X) must be imposed. The main strategies studied in this course are **instrumental variables** (Chapter 7), the **front-door criterion** (Chapter 3), and **sensitivity analysis**.

Table 1.1: The two scenarios arising from the back-door adjustment proposition.

	Scenario 1: U observed	**Scenario 2: U unobserved**
Confounding type	No unmeasured confounding	Unmeasured confounding
Back-door formula usable?	Yes	No
Identification route	Back-door / standardization	IV, front-door, …
Key assumption	Unconfoundedness: $U \perp\!\!\!\perp T \mid X$	Exclusion restriction, monotonicity, …
Chapters	2–6	7

1.4 The Gaussian Linear Confounded Model

1.4.1 Setup

The Gaussian linear confounded model is our primary running example throughout this course. It is simple enough to permit closed-form calculations yet rich enough to illustrate every key concept.

Example: Gaussian Linear Confounded Model

The structural equations are

$$Y = \beta T + \gamma U + \varepsilon, \qquad T = \alpha U + \delta, \tag{1.4}$$

where U is an unobserved confounder and the three primitive variables are mutually independent:

$$U \sim \mathcal{N}(0, \sigma_U^2), \qquad \varepsilon \sim \mathcal{N}(0, \sigma_\varepsilon^2), \qquad \delta \sim \mathcal{N}(0, \sigma_\delta^2).$$

Here β is the causal effect of T on Y, γ is the direct effect of the confounder U on Y, and α is the direct effect of U on T.

The endogeneity of T is immediate. From the analyst's perspective, the composite error in the Y equation is $\gamma U + \varepsilon$, so:

$$\mathrm{Cov}(T,\ \gamma U + \varepsilon) = \gamma\,\mathrm{Cov}(\alpha U + \delta,\ U) = \gamma\alpha\sigma_U^2 \ \neq\ 0 \quad (\text{when } \alpha \neq 0 \text{ and } \gamma \neq 0).$$

Consequently, OLS regressing Y on T does not estimate β.

The causal structure is captured by the DAG pair in Figure 1.3. In the observational DAG, the back-door path $T \leftarrow U \rightarrow Y$ is open: unobserved confounder U causes both T and Y, making T and the composite error $\gamma U + \varepsilon$ correlated. Under the intervention do(T=t), all arrows into T are deleted (shown faded), severing the back-door path.

Figure 1.3: Graph surgery on the Gaussian confounded model. The faded arrow in the mutilated DAG has been severed by the intervention do(T=t). The back-door path $T \leftarrow U \rightarrow Y$ is eliminated; only the causal path $T \rightarrow Y$ remains active. The treatment node becomes a fixed constant (rectangle).

1.4.2 Two Explicit Densities

Write $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$ for the marginal variance of T.

Interventional density. Set $T = t$ externally. The back-door path through U is severed, so U is independent of T under the intervention and is drawn from its marginal $\mathcal{N}(0, \sigma_U^2)$. Then $Y \mid \mathrm{do}(T{=}t) = \beta t + \gamma U + \varepsilon$ with $U \perp\!\!\!\perp \varepsilon$. Hence

$$f(y \mid \mathrm{do}(T{=}t)) = \mathcal{N}\big(\beta t,\ \ \gamma^2\sigma_U^2 + \sigma_\varepsilon^2\big)\,. \tag{1.5}$$

Observational density. Observe $T = t$. Since Y and T are both linear combinations of the independent normals (U, ε, δ), the pair (Y, T) is jointly normal. Their covariance is $\mathrm{Cov}(Y, T) = \beta\sigma_T^2 + \gamma\alpha\sigma_U^2$. Applying the conditional normal formula:

$$f(y \mid T{=}t) = \mathcal{N}\left(\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right)t,\ \ \frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2\right). \tag{1.6}$$

1.4.3 The Endogeneity Gap

Equations Equation 1.5 and Equation 1.6 differ in both mean and variance:

Table 1.2: *Equal when $\alpha = 0$ or $\gamma = 0$ (no confounding)*

	Interventional	**Observational**
Mean	βt	$\left(\beta + \frac{\gamma\alpha\sigma_U^2}{\sigma_T^2}\right) t$
Variance	$\gamma^2\sigma_U^2 + \sigma_\varepsilon^2$	$\frac{\gamma^2\sigma_U^2\sigma_\delta^2}{\sigma_T^2} + \sigma_\varepsilon^2$
Residual	$(\gamma U{+}\varepsilon) \perp\!\!\!\perp T$	$(\gamma U{+}\varepsilon)$ correlated with T
Identified by	RCT (or IV, Chapter 7)	OLS

The *endogeneity bias* of OLS is $\gamma\alpha\sigma_U^2/\sigma_T^2$: the coefficient of T in the observational mean exceeds the causal coefficient β by this amount. Numerically, with $\alpha = \gamma = 1$ and $\sigma_U^2 = \sigma_\delta^2 = 1$:

$$\text{bias} = \frac{1 \times 1 \times 1}{1^2 \times 1 + 1} = \frac{1}{2} = 0.5.$$

1.4.4 Lab: Simulating the Two Densities

The analytical results of this section were verified by simulation using $n = 10{,}000$ draws from the structural equations with parameters $\beta = 2$, $\alpha = 1$, $\gamma = 1$, $\sigma_U = \sigma_\varepsilon = \sigma_\delta = 1$. The R code is distributed separately as `chapter1_lab.R`.

Experiment 1: OLS bias. By the omitted-variable bias formula, the OLS probability limit is $\beta + \gamma\alpha\sigma_U^2/\sigma_T^2 = 2 + 0.5 = 2.5$.

	Simulated	**Theory**	**Formula**
OLS slope	2.5147	2.5000	$\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$
True β	—	2.0000	β
Bias	0.5147	0.5000	$\gamma\alpha\sigma_U^2/\sigma_T^2$

OLS overestimates the causal effect by 25%.

Experiment 2: Two-sample comparison at $t_0 = 1$. Theoretical values are from Equation 1.5 and Equation 1.6 at $t_0 = 1$.

	Mean (Simulated)	**Mean (Theory)**	**SD (Simulated)**	**SD (Theory)**
$\mathbb{E}[Y \mid T{=}1]$	2.567	2.500	1.240	$\sqrt{1.5} = 1.225$
$\mathbb{E}[Y \mid \text{do}(T{=}1)]$	1.999	2.000	1.423	$\sqrt{2} = 1.414$

The variance reduction in the observational distribution reflects the ratio $\sigma_{\text{obs}}^2/\sigma_{\text{int}}^2 = 1.5/2 = 0.75$.

Experiment 3: Density overlay for varying α. Figure 1.4 shows the interventional density $\mathcal{N}(\beta t_0, \gamma^2\sigma_U^2 + \sigma_\varepsilon^2)$ (solid) and the observational density Equation 1.6 (dashed), both evaluated at $t_0 = 1$, for $\alpha \in \{0, 0.3, 0.7, 1.0\}$.

1.4.5 The IV DAG Versus the Mediation DAG

The IV DAG introduced above looks, at first glance, structurally similar to the mediation DAG that will appear in Chapter 8. The similarity is superficial; the conceptual difference is fundamental. Confusing the two is one of the most common errors in applied causal inference.

Figure 1.5 places the two DAGs side by side.

Table 1.3 makes the comparison systematic.

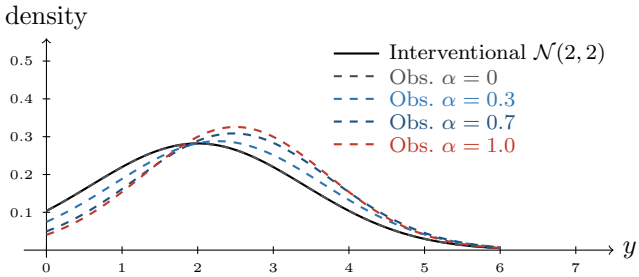

Figure 1.4: Interventional density $\mathcal{N}(2,2)$ (solid black) versus observational density for $\alpha \in \{0, 0.3, 0.7, 1.0\}$ (dashed), evaluated at $t_0 = 1$. As α increases, the observational curve shifts rightward (growing bias) and narrows (reduced variance). At $\alpha = 0$ there is no confounding and the two densities coincide. The interventional curve (solid) is invariant to α.

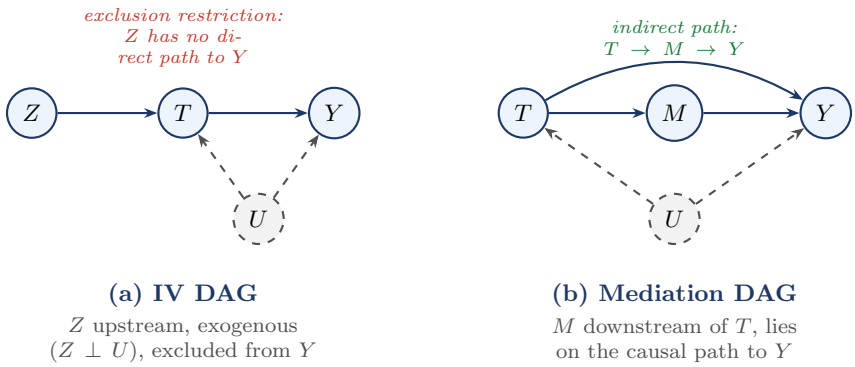

Figure 1.5: The IV DAG (a) and the mediation DAG (b) both involve a three-node chain, but they are structurally and conceptually distinct. In (a), the instrument Z is pre-treatment and exogenous: it satisfies $Z \perp\!\!\!\perp U$ and has no direct path to Y (exclusion restriction). In (b), the mediator M is post-treatment and endogenous: it is caused by T and transmits part of T's effect to Y.

Table 1.3: Instrumental variable versus mediator: a systematic comparison.

	Instrument Z	**Mediator** M
Position relative to T	Pre-treatment (upstream)	Post-treatment (downstream)
Causal relationship to T	Z causes T	T causes M
Direct path to Y?	**No** (exclusion restriction)	**Yes** ($M \to Y$ on causal path)
Independence from U?	**Yes** ($Z \perp\!\!\!\perp U$)	Not required
Primary purpose	Identify $T \to Y$ with unmeasured confounding	Decompose $T \to Y$ into direct and indirect
Key assumptions	Relevance, exogeneity, exclusion	Cross-world independence; no M–Y unmeasured confounding
Course coverage	Chapter 7	Chapter 8

The two chains look similar but are not interchangeable:

$$\underbrace{Z \longrightarrow T \longrightarrow Y}_{\text{IV chain: } Z \text{ is the identification device}} \qquad \text{versus} \qquad \underbrace{T \longrightarrow M \longrightarrow Y}_{\text{Mediation chain: } M \text{ is on the causal path}}$$

> Remark
>
> There is one important configuration where an observed post-treatment variable serves a role *analogous* to an instrument: the *front-door criterion* (Chapter 3). When $T \to M \to Y$ and $T \leftarrow U \to Y$ with U unobserved, but M is itself unconfounded, the causal effect of T on Y can still be identified by combining two back-door adjustments. This is the one case where a mediator also plays an identification role.

1.5 The Two-Step Paradigm

Causal inference is a two-step discipline. The two steps are logically distinct and require different tools.

1.5.1 Step 1: Identification

> Definition: Identification
>
> A causal quantity $P(y \mid \mathrm{do}(T{=}t))$ is *identified* from the observational distribution P if there exists a functional Ψ such that
> $$P(y \mid \mathrm{do}(T{=}t)) = \Psi(P),$$
> and $\Psi(P)$ depends only on the observable joint distribution.

Identification is a purely mathematical question about the structure of the causal model: given the assumed graph and assumptions, can the interventional density be expressed as a function of the observed-data distribution? It does not depend on sample size.

The main tools are: the back-door criterion (Chapters 2–3, applied in Chapters 5–6), the front-door criterion (Chapter 3), the three rules of the do-calculus (Chapter 3), and IV assumptions (Chapter 7).

1.5.2 Step 2: Estimation

Once $\Psi(P)$ has been identified, the statistical problem is to estimate $\Psi(P)$ from n observations $O_1, \dots, O_n$ as efficiently as possible. The main tools are efficient influence functions (EIF), inverse probability weighting (IPW), doubly robust estimators, and double machine learning (DML), all covered in Part III.

1.5.3 A Worked Example: Flu Vaccination and Infection

Scenario. A public health agency observes 1,000 individuals over one flu season. Each person either received a flu vaccine ($T = 1$) or did not ($T = 0$), and either became infected ($Y = 1$) or did not ($Y = 0$).

Age group X (elderly $= E$, young $= Y$) is recorded for all individuals. Elderly people are both more likely to be vaccinated and more susceptible to infection, so X confounds the relationship between T and Y.

The causal DAG is shown in Figure 1.6.

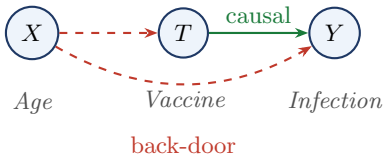

Figure 1.6: Causal DAG for the vaccination example. The green arrow is the causal path $T \to Y$. The red dashed arrows form the back-door path $T \leftarrow X \to Y$: age X causes both vaccination T and infection Y, creating a spurious association. Because X is observed, conditioning on it blocks the back-door path.

Observed data.

	Vaccinated ($T = 1$)		**Unvaccinated** ($T = 0$)		Total
	n	Infected	n	Infected	
Elderly ($X = E$)	360	108 (30%)	40	20 (50%)	400
Young ($X = Y$)	60	3 (5%)	540	54 (10%)	600
Total	420	111 (26.4%)	580	74 (12.8%)	1,000

Simpson's paradox. Within every stratum the vaccine reduces infection. Yet in the aggregate the vaccinated group has a *higher* infection rate (26.4% vs 12.8%). The reversal occurs because the vaccinated group is disproportionately elderly (86% of vaccinated vs. 7% of unvaccinated).

Stratum	Vaccinated	Unvaccinated	Difference	Conclusion
Elderly ($X = E$)	30%	50%	-20 pp	vaccine helps
Young ($X = Y$)	5%	10%	-5 pp	vaccine helps
Aggregate	**26.4%**	**12.8%**	$+13.6$ **pp**	**vaccine harms??**

Step 1: Identification. The set $\{X\}$ satisfies the back-door criterion: it blocks the path $T \leftarrow X \to Y$ and contains no descendant of T. By the back-door formula:

$$P(Y{=}1 \mid \mathrm{do}(T{=}t)) = \sum_{x \in \{E,Y\}} P(Y{=}1 \mid T{=}t, X{=}x)\, P(X{=}x). \tag{1.7}$$

Step 2: Estimation. Plugging sample proportions into Equation 1.7, with $\hat{P}(X{=}E) = 0.4$ and $\hat{P}(X{=}Y) = 0.6$:

$$\hat{P}(Y{=}1 \mid \mathrm{do}(T{=}1)) = 0.30 \times 0.4 + 0.05 \times 0.6 = 0.15, \qquad \hat{P}(Y{=}1 \mid \mathrm{do}(T{=}0)) = 0.50 \times 0.4 + 0.10 \times 0.6 = 0.26.$$

The estimated causal risk difference is $\hat{\Delta}_{\text{causal}} = 0.15 - 0.26 = -0.11$: the vaccine causally reduces infection probability by 11 percentage points.

The Two Steps, Concretely

Step 1 (identification) established — from the DAG alone — that the back-door formula (Equation 1.7) expresses the causal quantity as a functional of observable data.
Step 2 (estimation) replaced the population probabilities in that formula with sample proportions. These two steps are logically separate: Step 1 is a mathematical argument about the causal model that does not depend on sample size; Step 2 is a statistical argument about finite-sample estimation that does not depend on the causal model. A researcher who skips Step 1 and reports the raw difference conflates the two and draws the wrong conclusion.

1.6 Summary

1. **Interventional vs. observational distribution.** Causal inference answers questions about interventions $f(y \mid \mathrm{do}(T{=}t))$, not about observations $f(y \mid T{=}t)$. These two distributions coincide only under exogeneity; whenever T is endogenous they differ, and the difference is endogeneity bias.
2. **The causal trinity.** The same causal question can be expressed in three languages — SEM (equation surgery), DAG (graph surgery), and potential outcomes ($Y(t)$). The do-operator makes the interventional/observational distinction syntactically explicit. Fluency in all three is essential for modern causal reasoning.
3. **Endogeneity bias in the Gaussian confounded model.** In the Gaussian linear confounded model, the coefficient of T in the observational mean $\mathbb{E}[Y \mid T{=}t]$ exceeds the causal coefficient β by $\gamma\alpha\sigma_U^2/\sigma_T^2$, where $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$. This is the omitted-variable bias.
4. **Two-step paradigm.** Causal inference is a two-step discipline: *identification* (expressing $\Psi(P)$ as a functional of observable data, a mathematical question about the causal model) followed by *estimation* (constructing $\hat{\Psi}_n$ efficiently from n observations, a statistical question about finite samples). The two steps are logically independent.
5. **Instrument vs. mediator.** The IV DAG ($Z \to T \to Y$ with $Z \perp\!\!\!\perp U$) and the mediation DAG ($T \to M \to Y$) are structurally distinct. Conflating the two leads to violations of the exclusion restriction and identification failure.

Causal inference is the study of when the observational distribution contains enough information to recover the interventional distribution. Everything in this course — the back-door criterion, do-calculus, instrumental variables, doubly robust estimation — is an answer to that question in a particular setting.

1.7 Problems

1. Do-notation fundamentals. For each statement below, decide whether it refers to an interventional or an observational distribution, and rewrite it unambiguously using do-notation where appropriate.

(a) "The probability that a patient recovers given that they took the drug."
(b) "The probability that a patient would recover if we prescribed the drug to everyone."
(c) "Among students who attended tutoring, the average exam score was 85."
(d) "If we enrolled all students in tutoring, the average exam score would be 85."

2. Deriving the observational density. In the Gaussian confounded model, verify Equation 1.6 by carrying out the following steps. Let $\sigma_T^2 = \alpha^2\sigma_U^2 + \sigma_\delta^2$.

(a) Show that (Y, T) is jointly normal by writing both as linear combinations of the independent normals (U, ε, δ).
(b) Compute $\mathrm{Cov}(Y, T)$ and $\mathrm{Var}(T)$.
(c) Apply the conditional normal formula to derive $\mathbb{E}[Y \mid T{=}t]$ and $\mathrm{Var}[Y \mid T{=}t]$, and hence verify Equation 1.6.
(d) Show that the OLS probability limit is $\beta + \gamma\alpha\sigma_U^2/\sigma_T^2$, and interpret this as an omitted-variable bias formula.

3. The causal trinity. Consider the following verbal causal claim: "Aspirin (T) reduces the risk of heart attack (Y) because it inhibits platelet aggregation (M), but patients with pre-existing cardiovascular disease (U, unobserved) are both more likely to take aspirin and more likely to have a heart attack."

(a) Draw the causal DAG implied by this description.
(b) Write down the recursive SEM with appropriate structural functions f_T, f_M, f_Y.
(c) State the SUTVA assumption and discuss whether it is plausible in this setting.
(d) Explain why $\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$ does not identify the causal effect $\mathbb{E}[Y \mid \mathrm{do}(T{=}1)] - \mathbb{E}[Y \mid \mathrm{do}(T{=}0)]$ in this DAG.

4. IV versus mediation. For each scenario below, determine whether the intermediate variable described is an *instrument* or a *mediator*, and state which identifying assumption it satisfies or violates.

(a) A researcher uses distance to the nearest college (Z) to study the effect of college attendance (T) on earnings (Y).

(b) A nutritionist studies how a high-fiber diet (T) reduces cholesterol (M), which in turn reduces heart disease risk (Y).
(c) A political scientist uses rainfall on election day (Z) to estimate the effect of voter turnout (T) on electoral outcomes (Y).
(d) A pharmacologist studies whether a drug (T) reduces blood pressure (Y) by inhibiting a specific enzyme (M), and proposes to use M as an instrument for T. Evaluate this proposal.

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Chapter 2

DAGs and d-Separation

Learning Objectives

By the end of this chapter, students should be able to:

1. Construct a DAG from a verbal description of a causal model and identify parents, descendants, ancestors, and non-descendants of any node.
2. Classify every intermediate node on a path as a chain, fork, or collider, and apply the blocking rule for each.
3. Apply the d-separation criterion to determine all conditional independence relationships implied by a DAG.
4. Recognize and avoid collider bias, including the case where a descendant of a collider is conditioned upon.
5. State the Markov factorization and use it to write the joint density of a DAG in terms of local conditional densities.
6. Interpret d-separation results as conditional independence assumptions on observed variables, and recognize their role as the graphical expression of causal assumptions.

2.1 Directed Acyclic Graphs

DAGs allow us to translate qualitative causal assumptions into quantitative statistical restrictions. This single idea underlies everything in this chapter: once we draw a graph encoding what we believe about the causal structure of a problem, the graph tells us exactly which variables are independent, which are confounded, and what we need to condition on to identify a causal effect.

Remark

In this chapter, the word *graph* does not mean a plot of data or a graph of a function. It means a collection of nodes and edges used to represent relationships among variables. Our objective is to understand how such graphs encode statistical structure — especially conditional independence, confounding, and path blocking — and how this graphical structure will later support causal identification.

2.1.1 Basic Definition

Definition: Directed Acyclic Graph

A *directed acyclic graph* (DAG) $\mathcal{G} = (V, E)$ consists of a finite set of nodes V and a set of directed edges $E \subseteq V \times V$ such that there is no directed cycle: no node V_i has a directed path to itself.

In a causal DAG each node represents a variable and each directed edge $A \to B$ encodes "A is a direct cause of B, given all other variables in the model." The acyclicity condition is essential: it guarantees that

the variables can be ordered topologically so that the joint density admits the recursive factorization

$$p(v_1, \dots, v_k) = \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i)).$$

2.1.2 Structural Relationships

Definition: Structural Relationships in a DAG

For a node $v \in \mathcal{V}$ in $\mathcal{G} = (\mathcal{V}, E)$:

- $\mathrm{Pa}(v) = \{w \in \mathcal{V} : w \to v \in E\}$ are the *parents* of v.
- $\mathrm{De}(v)$ = all nodes reachable from v by directed paths are the *descendants* of v.
- $\mathrm{An}(v)$ = all nodes with a directed path to v are the *ancestors* of v.
- $\mathrm{Nd}(v) = \mathcal{V} \ (\mathrm{De}(v) \cup \{v\})$ are the *non-descendants* of v.

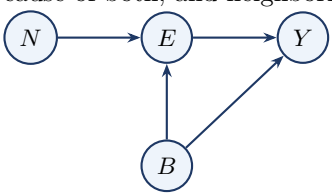
Note that $\mathrm{Nd}(v)$ includes the parents of v; when stating the local Markov property in Section 2.4, we will exclude the parents explicitly.

Quick Terminology Summary

A *node* represents a variable, and an *edge* represents a relationship between two variables. If $A \to B$, then A is a *parent* of B and B is a *child* of A. Two nodes are *adjacent* if they are connected by an edge. A *path* is any sequence of connected nodes, regardless of arrow direction; a *directed path* is a path whose arrows all point in the same direction. A node is an *ancestor* of another if there is a directed path from the former to the latter, and a *descendant* if the direction is reversed. A *directed acyclic graph* (DAG) is a directed graph with no directed cycles.

Example: Education, Earnings, and Family Background

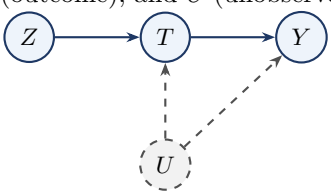
Consider the causal story: education (E) affects earnings (Y); family background (B) is a common cause of both; and neighborhood (N) affects education but has no direct effect on earnings.



The Markov factorization is $p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b)$. Reading off structural relationships: $\mathrm{Pa}(E) = \{N, B\}$, $\mathrm{Pa}(Y) = \{E, B\}$, $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$. The descendants of N are $\{E, Y\}$; B and N are non-descendants of each other.

Example: The IV DAG

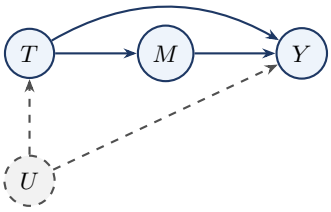
The canonical instrumental variable model involves four nodes: Z (instrument), T (treatment), Y (outcome), and U (unobserved confounder).



$\mathrm{Pa}(T) = \{Z, U\}$, $\mathrm{Pa}(Y) = \{T, U\}$, $\mathrm{Pa}(Z) = \mathrm{Pa}(U) = \emptyset$. The descendants of Z are $\{T, Y\}$; U and Z are non-descendants of each other.

Example: The Mediation DAG

In a mediation model, treatment T affects outcome Y both directly and through an intermediate variable M (the mediator). An unobserved confounder U creates a back-door path $T \leftarrow U \rightarrow Y$, making T endogenous. The mediator M is itself unconfounded.



$\mathrm{Pa}(T) = \{U\}$, $\mathrm{Pa}(M) = \{T\}$, $\mathrm{Pa}(Y) = \{M, T, U\}$, $\mathrm{Pa}(U) = \emptyset$. The total effect of T on Y travels along two paths: the direct path $T \rightarrow Y$ and the indirect path $T \rightarrow M \rightarrow Y$.

2.2 Paths, Blocking, and d-Separation

A DAG encodes direct causal relationships through its edges, but variables can also be statistically related through longer routes that mix directed and reversed edges. The concept of a *path* formalises these routes, and the question of whether a path transmits or blocks statistical dependence — depending on which variables are conditioned upon — is precisely what d-separation answers. The key insight is that conditioning can both *close* channels that were open (by blocking a chain or fork) and *open* channels that were closed (by activating a collider). Figure 2.1 illustrates all three cases.

2.2.1 Path Structures

Definition: Path

A *path* between nodes X and Y in $\mathcal{G}$ is any sequence of distinct nodes $X = V_0, V_1, \dots, V_k = Y$ such that each consecutive pair is connected by an edge in either direction.

Every intermediate node V_m on a path plays one of three structural roles. In a **chain** ($V_{m-1} \rightarrow V_m \rightarrow V_{m+1}$), V_m is a causal intermediary through which influence is transmitted. In a **fork** ($V_{m-1} \leftarrow V_m \rightarrow V_{m+1}$), V_m is a common cause that creates confounding between the two endpoints. In a **collider** ($V_{m-1} \rightarrow V_m \leftarrow V_{m+1}$), both arrows point *into* V_m; this is the crucial asymmetric case whose behavior under conditioning is the opposite of the other two.

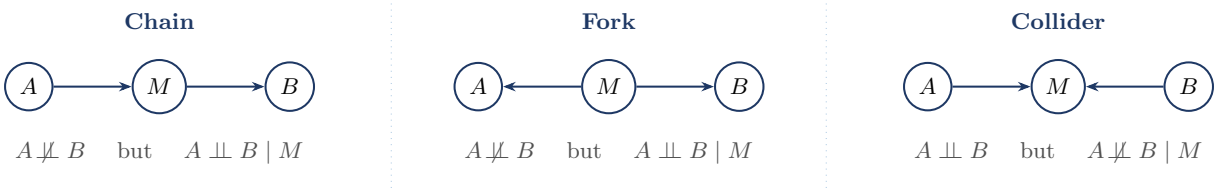


Figure 2.1: The three fundamental path structures in a DAG, with their default independence status and the effect of conditioning on the middle node M. Chains and forks are *open by default* and are blocked by conditioning on M. Colliders are *closed by default* and are opened by conditioning on M (or any descendant of M).

2.2.2 Blocking Rules

Conditioning on a node affects the paths it lies on differently depending on its structural role:

Structure	Pattern	Effect of conditioning	Key intuition
Chain	$A \to M \to B$	Blocks the path	Causal transmission
Fork	$A \leftarrow M \to B$	Blocks the path	Common cause
Collider	$A \to M \leftarrow B$	**Opens** the path	Common effect

The critical asymmetry — colliders are *closed by default* and *opened* by conditioning, while chains and forks are *open by default* and *closed* by conditioning — is what makes collider bias so easy to overlook in practice.

Chain: Smoking → Tar → Cancer. Let A = smoking, M = tar deposits, B = lung cancer. Marginally, heavy smokers have elevated cancer rates — the path is open. But conditioning on tar level blocks the path: once tar level is fixed, the causal channel is fully accounted for.

Fork: Poverty → Poor Diet; Poverty → Lack of Exercise. Let M = poverty, A = poor diet, B = lack of exercise. Unconditionally, diet quality and exercise are correlated through poverty. Conditioning on income level blocks the path and removes this spurious association.

Collider: Accident → Hospitalization ← Cancer. Let A = traffic accident, B = cancer, M = hospitalization. In the general population, $A \perp\!\!\!\perp B$. But among hospitalized patients — conditioning on M — the two become negatively associated. This is Berkson's bias (Berkson 1946).

Three-Node Motifs at a Glance

In a *chain* $A \to M \to B$, conditioning on M blocks the path. In a *fork* $A \leftarrow M \to B$, conditioning on M removes the spurious association. In a *collider* $A \to M \leftarrow B$, the path is blocked by default, but conditioning on M — or on any descendant of M — opens the path and induces association. These three motifs are the local building blocks of d-separation in arbitrary DAGs.

2.2.3 The d-Separation Criterion

Definition: d-Blocking and d-Separation

A path π is *d-blocked* by a set $\mathbf{S}$ if:
1. π contains a chain $A \to M \to B$ or fork $A \leftarrow M \to B$ with $M \in \mathbf{S}$; or
2. π contains a collider $A \to C \leftarrow B$ with $C \notin \mathbf{S}$ and no descendant of C is in $\mathbf{S}$.

Nodes X and Y are *d-separated* by $\mathbf{S}$, written $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$, if every path between X and Y in $\mathcal{G}$ is d-blocked by $\mathbf{S}$.

Example: Applying the d-Separation Definition to the Three Toy Settings

(i) Chain — Smoking → Tar → Cancer. The only path is $A \to M \to B$.
- $\mathbf{S} = \emptyset$: not d-blocked. $\Rightarrow A \not\perp\!\!\!\perp B$.
- $\mathbf{S} = \{M\}$: clause (1) applies. $\Rightarrow A \perp\!\!\!\perp B \mid M$.

(ii) Fork — Poverty → Poor Diet; Poverty → Lack of Exercise. The only path is $A \leftarrow M \to B$.
- $\mathbf{S} = \emptyset$: not d-blocked. $\Rightarrow A \not\perp\!\!\!\perp B$.
- $\mathbf{S} = \{M\}$: clause (1) applies. $\Rightarrow A \perp\!\!\!\perp B \mid M$.

(iii) Collider — Accident → Hospitalization ← Cancer. The only path is $A \to M \leftarrow B$.
- $\mathbf{S} = \emptyset$: collider $M \notin \mathbf{S}$, clause (2) applies. $\Rightarrow A \perp\!\!\!\perp B$.
- $\mathbf{S} = \{M\}$: clause (2) no longer blocks. $\Rightarrow A \not\perp\!\!\!\perp B \mid M$ (Berkson's bias).

Example: A Direct Probability Proof for the Chain

Consider the chain $X_1 \to X_2 \to X_3$. By the Markov factorization, $p(x_1, x_2, x_3) = p(x_1)\,p(x_2 \mid$

$x_1)\, p(x_3 \mid x_2)$. Conditioning on $X_2 = x_2$:

$$p(x_1, x_3 \mid x_2) = \frac{p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)}{p(x_2)} = p(x_1 \mid x_2)\, p(x_3 \mid x_2).$$

Hence $X_1 \perp\!\!\!\perp X_3 \mid X_2$. The fork case $X_1 \leftarrow X_2 \rightarrow X_3$ follows by an identical calculation.

Theorem: Verma & Pearl (1988); @pearl2009causality, Theorem 1.2.4

If $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$, then $X \perp\!\!\!\perp Y \mid \mathbf{S}$ in every distribution that is Markov with respect to $\mathcal{G}$.

Remark: Soundness, Not Converse

?@thm-dsep states that d-separation implies conditional independence for every distribution Markov with respect to $\mathcal{G}$. The converse need not hold without additional assumptions such as *faithfulness*: a conditional independence may occur in the data because of special parameter values even when the corresponding nodes are not d-separated in the graph.

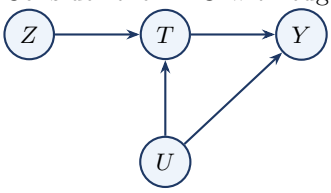
Table 2.1 summarizes the Markov factorization and its independence implication for each of the three path structures.

Table 2.1: Markov factorization and d-separation result for each path structure.

Structure	**Markov factorization** $p(a, m, b)$	**d-separation**
Chain $A \rightarrow M \rightarrow B$	$p(a)\, p(m \mid a)\, p(b \mid m)$	$(A \perp\!\!\!\perp B \mid M)_{\mathcal{G}}$
Fork $A \leftarrow M \rightarrow B$	$p(m)\, p(a \mid m)\, p(b \mid m)$	$(A \perp\!\!\!\perp B \mid M)_{\mathcal{G}}$
Collider $A \rightarrow M \leftarrow B$	$p(a)\, p(b)\, p(m \mid a, b)$	$(A \perp\!\!\!\perp B)_{\mathcal{G}}$

Example: Checking d-Separation in a Four-Node DAG

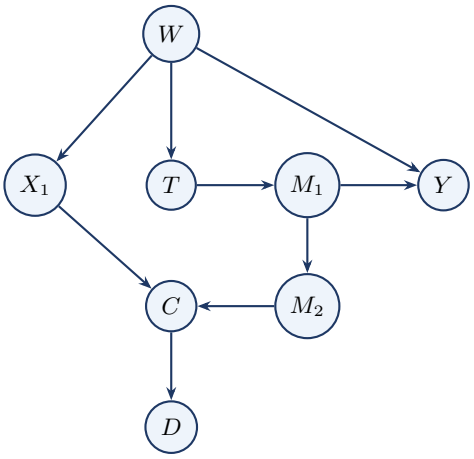
Consider the DAG with edges $Z \rightarrow T$, $U \rightarrow T$, $T \rightarrow Y$, $U \rightarrow Y$.



Query 1. Is $(Z \perp\!\!\!\perp U)_{\mathcal{G}}$? The only path $Z \rightarrow T \leftarrow U$ has T as a collider with $T \notin \emptyset$ — blocked. $\Rightarrow (Z \perp\!\!\!\perp U)_{\mathcal{G}}$.
Query 2. Is $(Z \perp\!\!\!\perp U \mid T)_{\mathcal{G}}$? Now $T \in \mathbf{S}$: the collider is observed, so the ball passes through. The path is open. $\Rightarrow Z \not\perp\!\!\!\perp U \mid T$.
Comparing the two queries: Z and U are marginally independent (the instrument is exogenous) but become dependent once we condition on T. This is Berkson's bias.

Example: Practice DAG — Eight-Node Graph

Consider the DAG with nodes W, X_1, T, M_1, M_2, Y, C, D and edges $W \rightarrow X_1$, $W \rightarrow T$, $W \rightarrow Y$, $T \rightarrow M_1$, $M_1 \rightarrow Y$, $M_1 \rightarrow M_2$, $X_1 \rightarrow C$, $M_2 \rightarrow C$, $C \rightarrow D$.



(1) Is $X_1 \perp\!\!\!\perp Y$? No. Two open paths: $X_1 \leftarrow W \rightarrow Y$ and $X_1 \leftarrow W \rightarrow T \rightarrow M_1 \rightarrow Y$.
(2) Is $X_1 \perp\!\!\!\perp Y \mid W$? Yes. Both open paths pass through the fork W; conditioning on W blocks them.
(3) Is $T \perp\!\!\!\perp M_2 \mid M_1$? Yes. The path $T \rightarrow M_1 \rightarrow M_2$ is blocked by M_1; all other paths are also blocked.
(4) Does conditioning on C create collider bias? Yes. C is a collider on $X_1 \rightarrow C \leftarrow M_2$. Conditioning on C opens this path, inducing a spurious association between X_1 and M_2.
(5) Does conditioning on D open the path $X_1 \rightarrow C \leftarrow M_2$? Yes. D is a descendant of C. By clause (2) of the d-separation definition, the collider at C is activated.

2.2.4 Practical Ways to Check d-Separation

Bayes-Ball Intuition. Imagine releasing a ball from X asking whether it can reach Y, given that nodes in $\mathbf{S}$ are observed. At a chain or fork node not in $\mathbf{S}$: ball passes through. At a chain or fork node in $\mathbf{S}$: ball stops. At a collider not in $\mathbf{S}$ (and no descendant in $\mathbf{S}$): ball stops. At a collider in $\mathbf{S}$ (or with a descendant in $\mathbf{S}$): ball passes through. The algorithm is formalized in Shachter (1998).

Moral Graph Transformation. (1) Restrict to the ancestral set $\mathrm{An}(X \cup Y \cup \mathbf{S})$. (2) Moralise: for every collider $A \rightarrow C \leftarrow B$, add an undirected edge $A - B$. (3) Remove all edge directions. (4) Delete all nodes in $\mathbf{S}$. If X and Y are disconnected in the result, then $(X \perp\!\!\!\perp Y \mid \mathbf{S})_{\mathcal{G}}$. A full treatment is in Lauritzen (1996, Ch. 3).

2.3 Collider Bias and the IV DAG

2.3.1 Berkson's Bias

Warning: Collider Bias [@berkson1946limitations]

Conditioning on a collider variable induces a spurious association between its parents, *even when they are marginally independent*. The same effect occurs when conditioning on any descendant of a collider.

Example: Collider Bias in the IV DAG

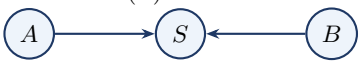
In the IV DAG, T is a collider on the path $Z \rightarrow T \leftarrow U$. Marginally, $Z \perp\!\!\!\perp U$ (the instrument is exogenous). However,

$$Z \perp\!\!\!\perp U \qquad \text{but} \qquad Z \not\perp\!\!\!\perp U \mid T.$$

Conditioning on T — restricting to the treated subpopulation — makes the instrument Z appear associated with U, invalidating the IV argument within that stratum.

Example: Collider Bias through Selection: Talent and Wealth

Suppose both talent (A) and wealth (B) increase the probability of admission to an elite school, so admission (S) is a collider: $A \to S \leftarrow B$.



In the general population, talent and wealth may be independent: $A \perp\!\!\!\perp B$. But among admitted students — conditioning on S — lower talent makes higher wealth more likely, and vice versa. This is collider bias: admission is a common effect of talent and wealth, and restricting to admitted students is precisely conditioning on that common effect.

2.3.2 Full d-Separation Analysis of the IV DAG

We work through the IV DAG with edges $Z \to T$, $T \to Y$, $U \to T$, $U \to Y$, where U is unobserved.

Endpoints	**Path**	**Type**	**Why**
$Z \leftrightarrow Y$	$Z \to T \to Y$	*Causal*	Directed from Z to Y; carries the IV signal
$Z \leftrightarrow Y$	$Z \to T \leftarrow U \to Y$	*Associational*	Collider at T; activates back-door via U
$Z \leftrightarrow U$	$Z \to T \leftarrow U$	*Associational*	Collider at T; dormant unless T conditioned on

Question 1. Is $Z \perp\!\!\!\perp Y \mid \emptyset$? Causal path $Z \to T \to Y$: open. Associational path: blocked by collider at T. $\Rightarrow Z \not\perp\!\!\!\perp Y$.

Question 2. Is $Z \perp\!\!\!\perp Y \mid T$? Causal path: blocked by T. Associational path: collider at T opened. $\Rightarrow Z \not\perp\!\!\!\perp Y \mid T$. The residual association is entirely spurious — a collider artefact, not a causal effect.

Question 3. Is $Z \perp\!\!\!\perp Y \mid \{T, U\}$? Both paths blocked. $\Rightarrow Z \perp\!\!\!\perp Y \mid \{T, U\}$.

Question 4. Is $Z \perp\!\!\!\perp U \mid \emptyset$? The only path $Z \to T \leftarrow U$ is blocked by collider at T. $\Rightarrow Z \perp\!\!\!\perp U$. This is *exogeneity*.

The four questions give a complete graphical account of the three IV assumptions. Question 1 establishes *relevance*; Question 4 establishes *exogeneity*; Question 3 establishes the *exclusion restriction*. Question 2 is the warning: conditioning on T alone simultaneously closes the causal channel and opens the confounded channel.

2.4 The Markov Property and Factorization

Proposition: Conditional Independence in the Three-Node Chain

Let $X_1 \to X_2 \to X_3$ be a chain with Markov factorization $p(x_1, x_2, x_3) = p(x_1)\, p(x_2 \mid x_1)\, p(x_3 \mid x_2)$. Then $X_1 \perp\!\!\!\perp X_3 \mid X_2$. The analogous result for the fork $X_1 \leftarrow X_2 \to X_3$ is left as an exercise.

Definition: Local Markov Property

A distribution P satisfies the *local Markov property* with respect to $\mathcal{G}$ if, for every node $V_i \in \mathcal{V}$:

$$V_i \perp\!\!\!\perp (\mathrm{Nd}(V_i)\ \mathrm{Pa}(V_i)) \mid \mathrm{Pa}(V_i).$$

That is, once we condition on the direct causes of a node, that node is independent of all variables that are neither its descendants nor its parents.

Remark

The *global Markov property* is the statement that every d-separation in $\mathcal{G}$ implies a conditional independence in P — precisely the content of **?@thm-dsep**. For DAGs, the local and global Markov properties are equivalent (Lauritzen 1996, Proposition 3.27).

An immediate consequence is the *Markov factorization*:

$$p(v_1, \dots, v_k) \;=\; \prod_{i=1}^{k} p(v_i \mid \mathrm{Pa}(v_i))\,. \tag{2.1}$$

Example: Markov Factorization in the Education Example

For the DAG with edges $N \to E$, $B \to E$, $B \to Y$, $E \to Y$:

$$p(n, b, e, y) = p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

A regression of Y on E alone is confounded by B. Conditioning on B blocks the back-door path $E \leftarrow B \to Y$ and identifies the causal effect $P(y \mid \mathrm{do}(E{=}e))$ via the back-door formula (Chapter 3).

Example: Markov Factorization in the IV DAG

Including the latent U:

$$p(z, t, y, u) = p(z)\, p(u)\, p(t \mid z, u)\, p(y \mid t, u).$$

Since U is unobserved, the observable factorization is obtained by marginalizing:

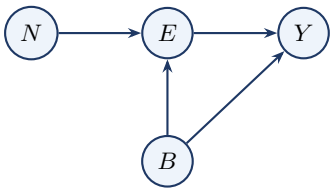
$$p(z, t, y) = \int p(z)\, p(u)\, p(t \mid z, u)\, p(y \mid t, u)\, \mathrm{d}u.$$

This cannot be simplified to $p(z)\, p(t \mid z)\, p(y \mid t)$ when U is a common cause of T and Y: this is precisely the endogeneity problem.

2.5 Worked Example: The Education–Earnings DAG

This section is the template for how we will use DAGs throughout the course: specify the DAG, read off the factorization, use d-separation to identify implied conditional independences, and interpret the result in causal terms.

The causal story. Education (E) affects earnings (Y); family background (B) is a common cause of both education and earnings; neighborhood (N) affects education but has no direct effect on earnings. All four variables are observed; there are no latent confounders.



Step 1 — Markov factorization. $\mathrm{Pa}(N) = \mathrm{Pa}(B) = \emptyset$, $\mathrm{Pa}(E) = \{N, B\}$, $\mathrm{Pa}(Y) = \{E, B\}$.

$$p(n, b, e, y) \;=\; p(n)\, p(b)\, p(e \mid n, b)\, p(y \mid e, b).$$

Step 2 — d-Separation. There are two paths between N and Y: Path 1: $N \to E \to Y$ (chain through E). Path 2: $N \to E \leftarrow B \to Y$ (collider at E).

- *Query (i):* $(N \perp\!\!\!\perp Y)_{\mathcal{G}}$? Path 1 open. $\Rightarrow N \not\perp\!\!\!\perp Y$.

- *Query (ii):* $(N \perp\!\!\!\perp Y \mid E)_{\mathcal{G}}$? Path 1 blocked by E; Path 2 collider E opened. Path still open through B. $\Rightarrow N \not\perp\!\!\!\perp Y \mid E$. This is a collider trap: conditioning on E inadvertently opens a spurious association between N and Y through B.
- *Query (iii):* $(N \perp\!\!\!\perp Y \mid \{E, B\})_{\mathcal{G}}$? Path 1 blocked by E; Path 2 collider opened but B blocks it. $\Rightarrow (N \perp\!\!\!\perp Y \mid E, B)_{\mathcal{G}}$.
- *Query (iv):* $(N \perp\!\!\!\perp B)_{\mathcal{G}}$? Path $N \to E \leftarrow B$: collider at E with $E \notin \emptyset$ — blocked. $\Rightarrow N \perp\!\!\!\perp B$.

Step 3 — Conditional Independence. By **?@thm-dsep**: $N \perp\!\!\!\perp B$, $N \perp\!\!\!\perp Y \mid E, B$, $N \not\perp\!\!\!\perp Y$, $N \not\perp\!\!\!\perp Y \mid E$.

Step 4 — Identification. The only back-door path is $E \leftarrow B \to Y$. With $\mathbf{S} = \{B\}$:

$$P(y \mid \mathrm{do}(E{=}e)) \;=\; \sum_{b} P(y \mid e, b)\, P(b).$$

Every term involves only the observed distribution, so the causal effect is identified entirely from observational data.

2.6 The Big Picture

The concepts developed in this chapter form the graphical foundation for all subsequent identification and estimation theory. The chain of reasoning runs from a qualitative causal graph all the way to a formula for the interventional distribution:

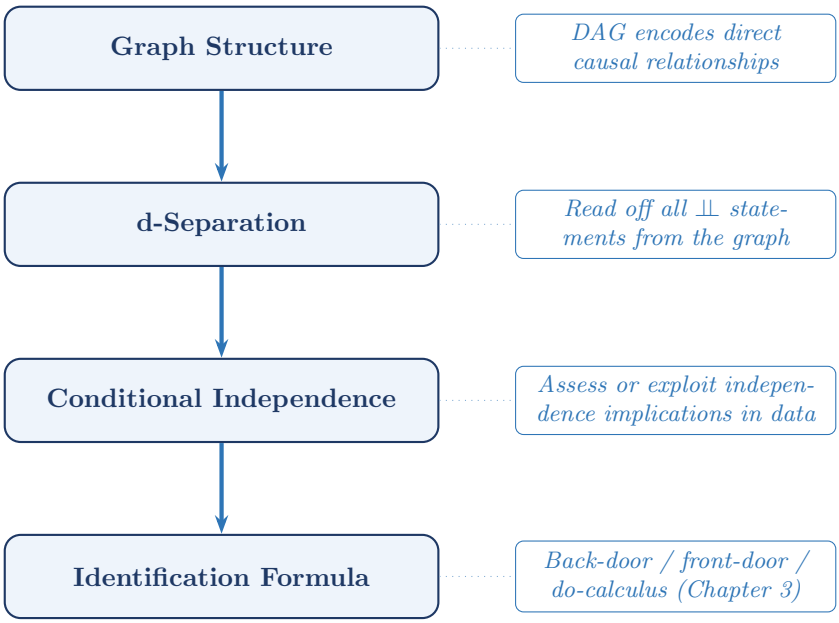


Figure 2.2: The four-step pipeline from causal assumptions to identification formula.

The logical chain in compact form:

structural assumptions $\Longrightarrow$ DAG $\Longrightarrow$ d-separation $\Longrightarrow$ conditional independences $\Longrightarrow$ identification formulas.

2.7 Summary

1. A DAG encodes the qualitative causal structure of a data-generating mechanism without committing to functional forms or error distributions.
2. The three elementary path patterns — chain, fork, and collider — determine when conditioning on a middle node blocks or opens a path. Chains and forks are *open by default* and blocked by conditioning; colliders are *closed by default* and opened by conditioning.
3. The d-separation criterion is a graphical rule: X and Y are d-separated by $\mathbf{S}$ exactly when every path between them is blocked by $\mathbf{S}$. Under the Markov property, this graphical statement implies a corresponding conditional independence in the distribution.

4. The Markov factorization $p = \prod_i p(x_i \mid \mathrm{Pa}(x_i))$ is the formal bridge between the graph and the joint distribution, and is the starting point for all identification derivations.

5. Collider bias (Berkson 1946) is one of the most common errors in applied causal inference. Conditioning on a descendant of a collider opens the collider just as conditioning on the collider itself does.

6. D-separation provides a graphical criterion for reading off conditional independence implications of a DAG. Under the Markov property, these imply corresponding conditional independences in the observed distribution. The IV DAG analysis illustrates this directly: the claim that Z is a valid instrument is encoded as d-separation statements in the graph, yet the core assumptions of exogeneity and exclusion are substantive and not fully verifiable from observational data alone.

7. **A practical workflow.** To analyze a DAG, proceed in four steps: identify the graph structure, determine which paths are open or blocked, translate d-separation statements into conditional independences under the Markov property, and then interpret those independences in light of the causal question.

2.8 Problems

1. Warm-up: a single collider. Consider the DAG $X \to Y \leftarrow Z$, where X and Z have no other connections.

(a) Identify the structural role of Y on the path $X \to Y \leftarrow Z$.
(b) Is $X \perp\!\!\!\perp Z$? Apply the d-separation criterion to the only path between X and Z.
(c) Is $X \perp\!\!\!\perp Z \mid Y$? Explain what happens to the path when Y is conditioned on.

2. d-Separation practice. Consider the DAG: $A \to B \to D$, $A \to C \to D$, $B \to E$, $C \to E$.

(a) List all paths between A and E. (*Hint*: there are four paths in total; two pass through D.)
(b) For each path, identify the role (chain, fork, collider) of each intermediate node.
(c) Does $\{B, C\}$ d-separate A and E?
(d) Does $\{D\}$ d-separate B and C? What type of node is D on the path $B \to D \leftarrow C$?

3. Berkson's bias. Suppose X and Y are independent standard normal variables, and let $S = \mathbf{1}[X + Y > 0]$.

(a) Verify analytically that $\mathrm{Cov}(X, Y \mid S{=}1) < 0$.
(b) Draw the DAG for (X, Y, S) and identify S as a collider.
(c) Explain in one sentence why restricting the analysis to the subsample with $S = 1$ biases estimates of any association between X and Y.

4. Markov factorization and collider activation. Consider the DAG with edges $A \to E$, $A \to W$, $F \to E$, $E \to W$, where $A =$ ability, $F =$ family income, $E =$ education, $W =$ wages.

(a) Write down the Markov factorization $p(a, f, e, w)$.
(b) Is $(F \perp\!\!\!\perp W)_{\mathcal{G}}$? List all paths between F and W.
(c) Is $(F \perp\!\!\!\perp W \mid E)_{\mathcal{G}}$? Identify the role of E on each path.
(d) A researcher regresses W on E and F, omitting A. Is the coefficient on E a causal effect? Explain using the graph.

5. Theorem proof for the collider. Consider the collider $A \to M \leftarrow B$ with Markov factorization $p(a, m, b) = p(a)\, p(b)\, p(m \mid a, b)$.

(a) By marginalizing over M, show that $A \perp\!\!\!\perp B$ in the joint distribution.
(b) Show that conditioning on M breaks this independence.
(c) Explain in one sentence why parts (a) and (b) together are consistent with **?@thm-dsep**.

6. Terminology check. Consider the DAG with edges $U \to X$, $U \to Z$, $X \to W$, $Z \to W$, $W \to Y$.

(a) Identify the parents, children, ancestors, descendants, and non-descendants of node W.
(b) List all pairs of adjacent nodes.
(c) Which pairs of nodes are connected by a directed path?
(d) Write the Markov factorization $p(u, x, z, w, y)$.

7. Markov factorization and local Markov property. Consider the DAG with edges $X_1 \to X_2$, $X_1 \to X_3$, $X_2 \to X_4$, $X_3 \to X_4$.

(a) Write the joint density $p(x_1, x_2, x_3, x_4)$ implied by the Markov factorization.
(b) State the local Markov property for each of the four nodes.
(c) Is $(X_2 \perp\!\!\!\perp X_3)_{\mathcal{G}}$? Is $(X_2 \perp\!\!\!\perp X_3 \mid X_1)_{\mathcal{G}}$?

8. Toy proof: conditional independence in the fork. Consider the fork $X_1 \leftarrow X_2 \to X_3$ with Markov factorization $p(x_1, x_2, x_3) = p(x_2)\, p(x_1 \mid x_2)\, p(x_3 \mid x_2)$.

(a) Show directly that $X_1 \perp\!\!\!\perp X_3 \mid X_2$.
(b) Is $X_1 \perp\!\!\!\perp X_3$ marginally?
(c) Explain why conditioning on the common cause X_2 removes the association between X_1 and X_3.

9. d-Separation in the practice DAG. Refer to the eight-node DAG in the practice example above. For each query, state whether it is true or false and justify by listing all relevant paths.

(a) $X_1 \perp\!\!\!\perp Y$
(b) $X_1 \perp\!\!\!\perp Y \mid W$
(c) $T \perp\!\!\!\perp M_2 \mid M_1$
(d) Does conditioning on C induce collider bias between X_1 and M_2?
(e) Does conditioning on D (a descendant of C) open the path $X_1 \to C \leftarrow M_2$?

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